Deep Residual Learning vs Vision Transformers for Visual Recognition

Abstract
Visual recognition architectures have evolved rapidly from convolutional networks like ResNet to attention-driven Vision Transformers (ViT). In this work, we evaluate ResNet-50 and ViT-Base on CIFAR-10 and ImageNet-1k datasets across 300 training epochs.

1. Introduction
Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. ResNet-50 achieved 85.3% top-1 accuracy on CIFAR-10.

2. Empirical Setup & Methodology
Both models were trained using AdamW optimizer with a base learning rate of 1e-3 and cosine learning rate decay over 300 epochs on Apple Silicon M4 Pro MPS acceleration.

3. Results & Convergence Analysis
Figure 3 demonstrates the top-1 accuracy curves across epochs. ResNet converges faster in early epochs (0-50), but ViT surpasses ResNet beyond epoch 120, reaching 88.1% top-1 accuracy.

4. Conclusion
Our experiments confirm that Vision Transformers demonstrate superior long-range dependency modeling when evaluated on large-scale vision benchmarks.